

Gravitation II

Assignment #1

Due: Esfand 15, 1404

Problem 1. Show that the number of linearly independent p -forms on an n -dimensional vector space is $\frac{n!}{n!(n-p)!}$.

Problem 2. Show that for any form A we have $d(dA) = 0$ where d takes the exterior derivative defined by $(dA)_{\mu_1\mu_2\ldots\mu_{p+1}} = (p+1)\partial_{[\mu_1}A_{\mu_2\ldots\mu_{p+1}]}$.

Problem 3. Consider the 3 Euclidean space (and therefore do not worry about the distinction between vectors and 1-forms). Assume A is a 1-form (or vector) in this space. Show that
a) the Hodge dual of the exterior derivative of A , i.e., $*(dA)$, would be a map from the space of 1-forms to 1-forms (or vectors to vectors), and is equivalent to $*dA = \nabla \times A$.
b) Similarly, show that $*d * A = \nabla \cdot A$.

Problem 4. a) Show that the homogeneous Maxwell equations (with no source) can be written concisely as $dF = 0$, where $F_{\mu\nu}$ is the 2-form representing the electromagnetic tensor.
b) Show that the non-homogeneous Maxwell equation can be compactly written as $d * F = *J$ where J is the current 1-form.